# LESSON PLAN

## EHM (28)

| | | | |
|---|---|---|---|
| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 07 |
| **Total No. of Lesson Plans** | 7 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

| | |
|---|---|
| **1. Topic** | Composite repair - procedure and safety precautions while working of composites. |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Composite repair - procedure and safety precautions while working of composites. |
| **5. Specific Objectives** | 28.1.3.5 Trainee will be able to explain Composite repair procedure. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

| | |
|---|---|
| **7. Practical in Lab/Equipment Rooms/Simulators** | Demonstration of composite repair procedure steps and safety gear usage; Instructor supervision of safety protocols. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

| | |
|---|---|
| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What are the primary hazards associated with composite materials?
(b) List the standard personal protective equipment required for composite work.
(c) Define the term 'delamination' in the context of composite structures.

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Safety precautions during composite handling<br>• Personal protective equipment requirements<br>• Ventilation and fire safety measures<br>• Handling of resin and hardener chemicals | L & Ds | PPT, IB | 15 min |
| (b) | • Surface preparation and cleaning procedures<br>• Lay-up techniques and resin application<br>• Curing processes and environmental controls<br>• Post-cure inspection and quality checks | L & Ds | PPT, PDF | 20 min |
| (c) | • Documentation of repair procedures<br>• Common defects and their identification<br>• Waste disposal of composite materials<br>• Final acceptance criteria for repairs | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the critical steps involved in the composite repair procedure.
(b) Describe the safety precautions necessary when working with composite resins.

| | |
|---|---|
| **12. Home Assignment and Follow Up** | Prepare a brief summary of the composite repair procedure to be discussed in the next lesson plan (LP 02). |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | End of subject | Refer to the master syllabus reference list | - |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min